Module 2: Donor integration, cell-class composition, and pathway analysis

Instructional time: 45 min | Bloom: Apply, Analyze | Notebook: notebooks/02_integration_de_pathways.ipynb

1. Overview

Module 2 stays inside HuBMAP. It concatenates the four Module 1 QC objects, asks whether Harmony improves donor mixing without erasing biology, maps coarse cell_class labels, separates composition change from expression change, and runs Wilcoxon marker detection plus local gene-set enrichment (Hallmark / Reactome / GO Biological Process).

Key components. Donor concatenation and Harmony; mixing metrics; composition tables and dual pseudobulk summaries; marker DE; ORA / prerank enrichment from shipped GMT files under data/genesets/.

Learning objectives.

- Integrate multi-donor snRNA-seq and quantify whether batch correction increases foreign-donor neighborhood mixing.
- Distinguish "is this population larger?" (composition) from "are these cells different?" (markers / pathways).
- Run and read local enrichment without treating top terms as automatically true biology.

Measured peak RSS on the DE leg sets the course RAM floor: **16 GB recommended, 8 GB minimum** (outputs/reports/module2_de_run_params.json).

2. Before you begin

Prerequisites. Module 1 (QC-preprocessed teaching blocks for Donor_1, Donor_2, Donor_3, Donor_4)

Time. Instructional: 45 min. Compute (measured): integration ~32 s, DE/pathways ~24 s; peak RSS ~7.6 GB on DE (16 GB recommended; 8 GB minimum).

Data required.

Run order. Run the analysis sections in order; the notebook carries state between cells.

Biological question

After QC, do the four HuBMAP lung donors form a usable joint embedding, and what cell-state and pathway differences appear across coarse lineages once donor identity is reduced?

Data used here

Input	Path pattern	Role
Module 1 QC objects	data/processed/hubmap/module1_*_qc_preprocessed.h5ad	Per-donor nuclei
Azimuth labels	cached TSV and/or remote range-read	cell_class mapping
Gene sets	data/genesets/*.gmt (~2.17 MB shipped)	Enrichment libraries

GTEx is intentionally absent until Module 4. Module 2 does not download new primary matrices beyond what Module 1 verified.

Cohort and ages are unchanged from Module 1 (Donor_1 to Donor_4). The age split remains n=2 vs n=2 illustration only.

3. Methods

Parameters below are live values under module2 in config/paths.yaml. Integration scripts are scripts/nb03_integration/; DE and enrichment are scripts/nb04_de/. Module 2 stays inside HuBMAP; GTEx is off (module2.gtex_context: false).

Each Methods step below corresponds to a numbered section of notebooks/02_integration_de_pathways.ipynb, so the document and the notebook can be read side by side:

Methods step	Notebook section
1. Load and concatenate Module 1 QC objects	1. Load, concatenate, and compute the joint embedding
2. Joint embedding (recomputed HVG / PCA)	1. Load, concatenate, and compute the joint embedding
3. Harmony batch correction	2. Harmony batch correction and mixing diagnostics
4. Mixing and silhouette diagnostics	2. Harmony batch correction and mixing diagnostics
5. Coarse cell_class mapping, composition, and pseudobulk	3. Coarse cell_class mapping; 4. Composition, markers, and weighted pseudobulk
6. Intron exclusion on the DE path, Wilcoxon markers	5. Differential expression and pathway enrichment
7. Local GMT enrichment (ORA and prerank)	5. Differential expression and pathway enrichment

Where two steps share a notebook section, one call performs both. Notebook section 0 is environment setup, covered in Module 0. Notebook section 6, composition versus expression, is reported under Results below rather than as a Methods step.

1. Load and concatenate Module 1 QC objects

Script. scripts/nb03_integration/concatenate.py (load_module2_inputs, concatenate_donors); path resolve via scripts/nb02_qc/export.py (resolve_qc_h5ad).

What. Load the four Module 1 QC-preprocessed AnnData objects listed in module2.inputs (post-QC, possibly subsampled teaching objects). Ensure donor metadata (donor_id, donor_label, block_id), uniquify observation names across donors, intersect on shared genes (prefer versionless Ensembl gene_id_no_version), and concatenate with an inner join. Outer-join fill is avoided so missing genes are not invented as zeros. **Intron-feature exclusion does not happen here.** Salmon intron-tagged features (*-I) remain in the integrated object until the DE load path (step 6) when module2.exclude_intron_features is true.

Parameters: see Appendix, section 1.

Why. A joint embedding requires a shared feature space built from the same QC policy. Concatenating Module 1 objects (not raw portal matrices) keeps filters and the teaching subsample explicit. Using post-QC subsampled objects means Module 2 composition is not interchangeable with the Module 1 pre-QC secondary_analysis airway gate.

2. Joint embedding (recomputed HVG / PCA)

Script. scripts/nb03_integration/integrate.py (recompute_pca_umap).

What. On the concatenated object, recompute highly variable genes, scale, and PCA, then neighbors and UMAP on X_pca (pre-Harmony view). Module 1 per-donor PCA/UMAP is **not** reused. Teaching defaults use n_pcs=40 (not Module 1's 50) so the joint basis is defined at integration time.

Parameters: see Appendix, section 2.

Why. Joint HVG/PCA puts all donors in one feature basis before batch correction. Reusing Module 1 embeddings would mix incompatible PCA spaces and hide the integration step learners are meant to inspect.

3. Harmony batch correction

Script. scripts/nb03_integration/integrate.py (run_harmony).

What. Run harmonypy on X_pca with batch key donor_id, store corrected embedding in obsm['X_pca_harmony'], then recompute neighbors (key harmony) and UMAP (X_umap_harmony) using the same embedding neighbor/UMAP knobs. theta defaults to null (harmonypy default); max_iter_harmony=20.

Parameters: see Appendix, section 3.

Why. Donor is a technical factor in multi-donor atlases. Harmony adjusts the PCA embedding so neighborhoods mix donors while preserving local structure better than regressing donor as a linear covariate in expression space. Mixing metrics (next step) are required; a prettier UMAP is not evidence.

4. Mixing and silhouette diagnostics

Script. scripts/nb03_integration/integrate.py (harmony_mixing_metrics, harmony_silhouette_by_celltype).

What. When module2.diagnostics.enabled is true, score mean foreign-donor kNN fraction on X_pca versus X_pca_harmony (k=knn_k, optional subsample to max_cells), and donor silhouette within coarse cell_class. These are learner diagnostics, not gates that block DE.

Parameters: see Appendix, section 4.

Why. Foreign-donor kNN rising after Harmony is the quantitative mixing claim. Within-class donor silhouette becoming more negative usually agrees. Read both with marker coherence so overmixing is visible.

5. Coarse cell_class mapping, composition, and pseudobulk

Script. scripts/nb03_integration/integrate.py (assign_coarse_cell_class, assert_azimuth_mapping_coverage, donor_composition, cell_class_membership_frame); scripts/nb03_integration/compare.py (composition_weighted_pseudobulk, max_across_labels_pseudobulk, label_fractions); export in scripts/nb03_integration/export.py.

What. Map Azimuth fine labels to **exactly four** coarse buckets: epithelial, endothelial, stromal, immune. Airway secretory and gland labels (Club, Goblet, SMG, Deuterosomal, transitional Club-AT2, Tuft, and related) fold into **epithelial**. There is no fifth coarse airway class. Both Unicode and ASCII macrophage keys map to immune (Azimuth Monocyte-derived M + Greek phi, and ASCII Monocyte-derived Mphi; likewise interstitial macrophage variants with phi / Mphi). Unmapped labels are listed loudly and never silently bucketed; unlabeled nuclei are dropped before DE group sizes are quoted.

Build donor-by-label crosstabs and cell-type fractions on the **post-QC subsampled** integrated object. When module2.export_pseudobulk is true, export composition-weighted and max-across-labels linear-scale pseudobulk vectors for Module 4's HuBMAP-GTEx bridge (plus fractions).

Parameters: see Appendix, section 5.

Why. Four-bucket labels keep classroom DE interpretable and force airway biology into epithelial rather than inventing a fifth DE group. Two pseudobulks answer different questions: composition-weighted tracks tissue-level bulk when abundances differ; max-across asks for the strongest class program. Module 4 compares them explicitly.

6. Intron exclusion on the DE path, Wilcoxon markers

Script. scripts/nb04_de/load.py (load_integrated, exclude_intron_features); scripts/nb04_de/markers.py (run_rank_genes, run_rank_genes_for_prerank, marker filtering helpers).

What. Load the integrated h5ad for DE. When module2.exclude_intron_features: true, drop Salmon intron-tagged features (*-I) **on this DE object only**; the on-disk integrated embedding object retains them unless rewritten. Record pre/post feature-space descriptions in adata.uns['module2_feature_space']. Run Wilcoxon rank-sum markers per cell_class (reference rest) with configured n_genes, min_cells_per_group, and later filter by min_logfoldchange / max_padj. Optionally exclude ribosomal genes from marker tables. A separate full-universe rank supports prerank GSEA.

Parameters: see Appendix, section 6.

Why. Intron exclusion is a DE-path decision so embedding feature space stays comparable to Module 1 gene intersection while marker ranks avoid intron-duplicate inflation. Wilcoxon on log-normalized X is the teaching default; counts remain in layers['counts'].

7. Local GMT enrichment (ORA and prerank)

Script. scripts/nb04_de/enrich.py (resolve_gmt_paths, run_enrichment_for_groups, run_prerank_gsea).

What. Over-representation enrichment per cell_class against local GMT files (Hallmark 2020, Reactome 2022, GO Biological Process 2023; ****), with background equal to genes measured in the DE object (enrichment.mode: local_gmt). Take top top_genes_per_group markers passing padj cutoff. Separately run preranked GSEA on Wilcoxon scores with min_size=15, max_size=500, permutation_num=100. Enrichr online remains opt-in only.

Parameters: see Appendix, section 7.

Why. Local GMTs keep teaching offline and license-clear. Enrichment is hypothesis-generating: inspect gene membership and n_input_genes before claiming a pathway story. Prefer terms that recur across libraries and that match known lung markers in the dotplots.

Production coarse group sizes on the integrated object are epithelial 6,928, endothelial 6,205, stromal 3,190 and immune 2,556 nuclei. A fifth group, unlabeled, holds 5 nuclei, falls below the min_cells_per_group floor of 50, and is dropped and reported rather than merged into a neighbor.

4. Results

Donor composition after concatenation

donor_label	donor_id	block_id	n_nuclei	n_genes	source_h5ad
Donor_1	HBM943.SCQQ.877	HBM347.VDHS.379	3884	45859	data/processed/hubmap/module1_HBM347.VDHS.379_qc_preprocessed.h5ad
Donor_2	HBM994.NXZC.854	HBM473.NKMR.872	5000	49754	data/processed/hubmap/module1_HBM473.NKMR.872_qc_preprocessed.h5ad

Donor_3	HBM643.KCCK.866	HBM484.RGWP.797	5000	53014	data/processed/hubmap/module1_HBM484.RGWP.797_qc_preprocessed.h5ad
Donor_4	HBM428.GRPJ.489	HBM379.QCJT.435	5000	47770	data/processed/hubmap/module1_HBM379.QCJT.435_qc_preprocessed.h5ad

Source: module2_donor_composition_summary.tsv

Concatenation keeps 18,884 nuclei: 3,884 from Donor_1 and 5,000 each from Donor_2, Donor_3, and Donor_4, at the module1.max_nuclei teaching cap. Coarse mapping sorts them into four uneven cell_class buckets: 6,928 epithelial, 6,205 endothelial, 3,190 stromal, and 2,556 immune. A fifth group of 5 unlabeled nuclei falls under the min_cells_per_group floor of 50 and is dropped, leaving 18,879 nuclei for DE.

Harmony mixing

embedding	k	n_cells_scored	mean_foreign_donor_knn_fraction	median_foreign_donor_knn_fraction
X_pca	30	8000	0.394337	0.4
X_pca_harmony	30	8000	0.685058	0.7

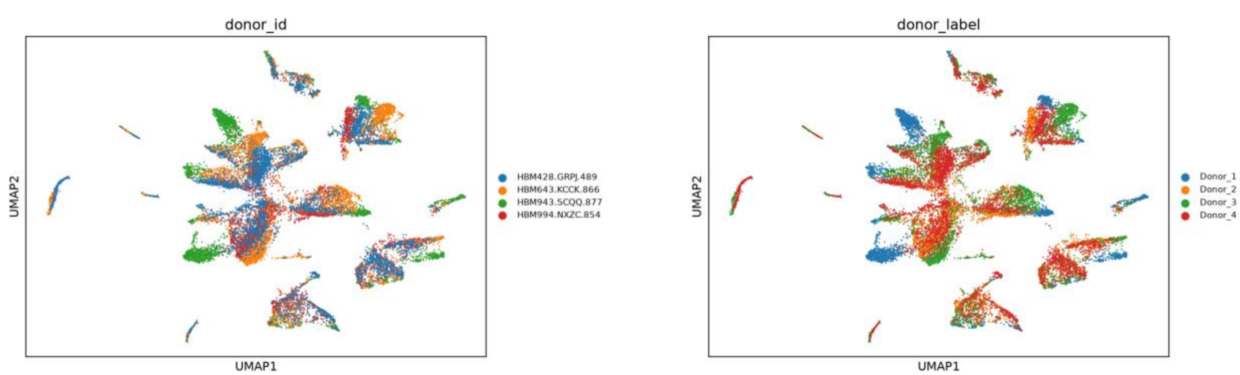
Source: module2_harmony_mixing_metrics.tsv

Mean foreign-donor kNN fraction rises from 0.394 on X_pca to 0.685 on X_pca_harmony at k=30 across 8,000 scored cells, and the median moves from 0.40 to 0.70. That is the mixing claim, and it is a global average.

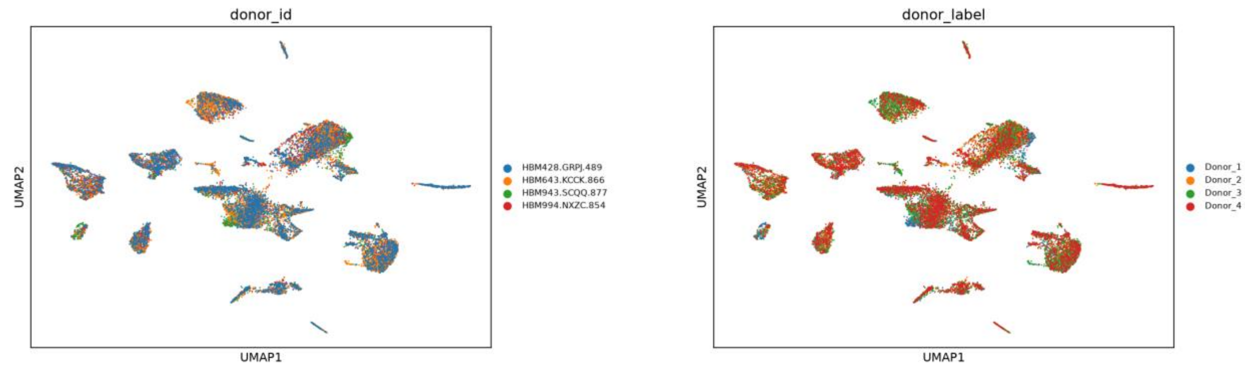
label	embedding	n_cells	n_donors	silhouette_donor
endothelial	X_pca	6205	4	0.004068
endothelial	X_pca_harmony	6205	4	-0.014194
epithelial	X_pca	6928	4	-0.007476
epithelial	X_pca_harmony	6928	4	-0.039982
immune	X_pca	2556	4	-0.029976
immune	X_pca_harmony	2556	4	-0.023069
stromal	X_pca	3190	4	0.011961
stromal	X_pca_harmony	3190	4	-0.027011

Source: module2_harmony_silhouette_by_celltype.tsv

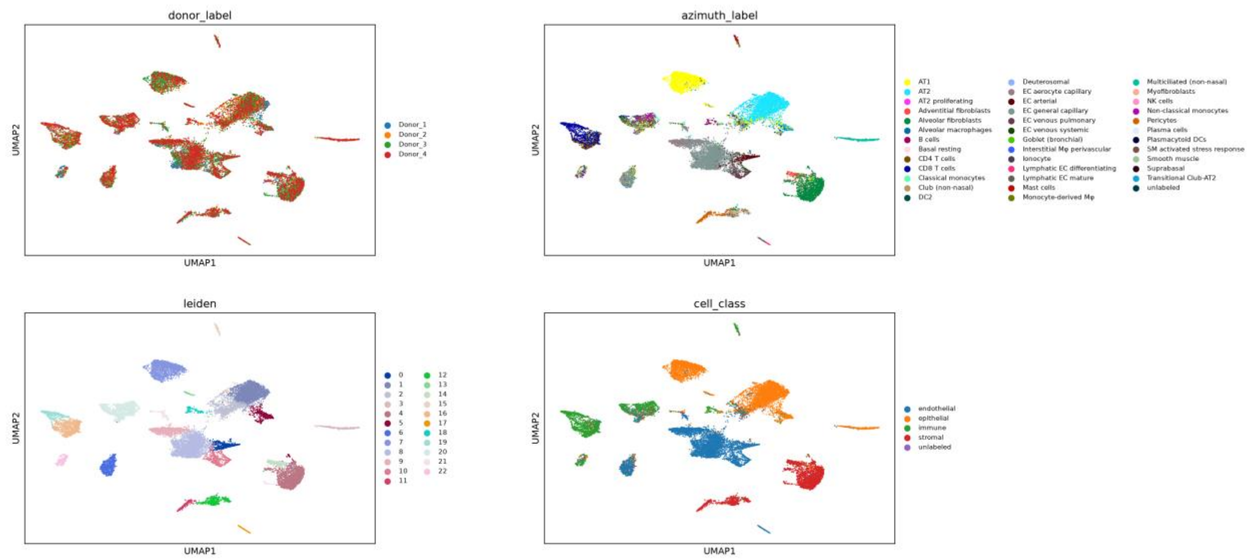
Per compartment, agreement is not unanimous. Donor silhouette falls after Harmony in three: endothelial 0.004 to -0.014, stromal 0.012 to -0.027, epithelial -0.007 to -0.040. Immune moves the opposite way, from -0.030 up to -0.023, toward zero. Immune is the smallest compartment and was already the best mixed before correction. Quote the exception rather than averaging it into the global number.



UMAP before Harmony colored by donor

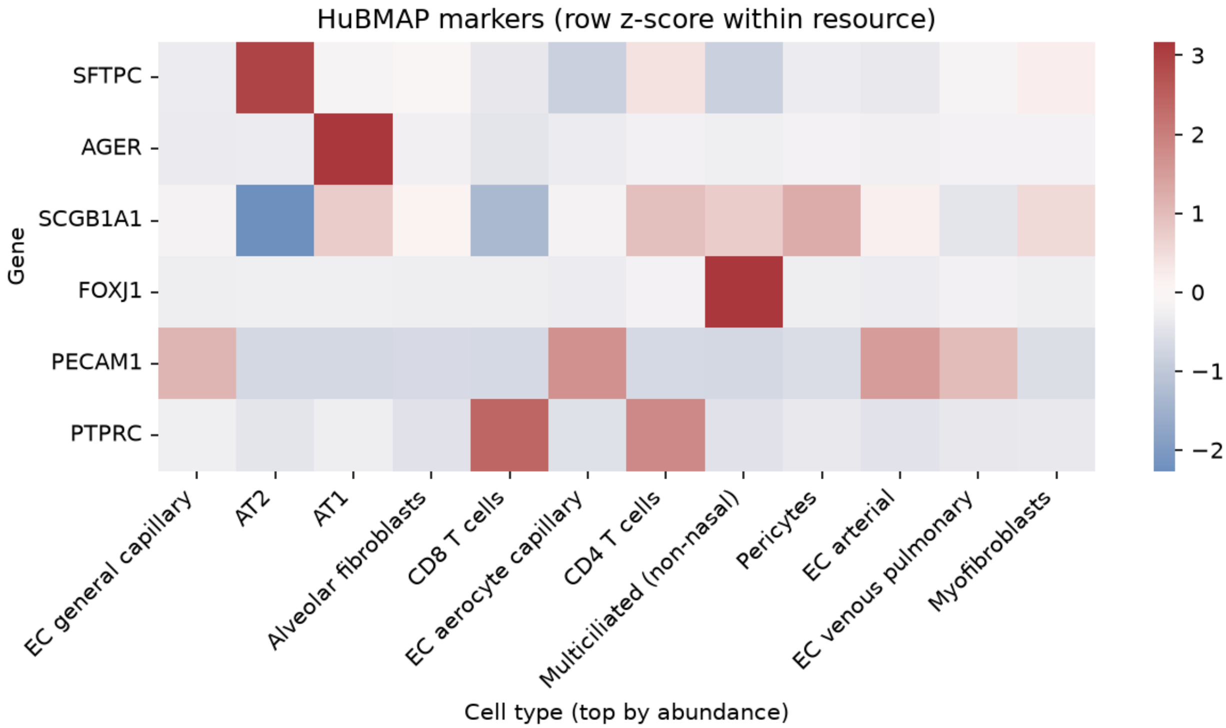


UMAP after Harmony colored by donor, the same view as the figure above

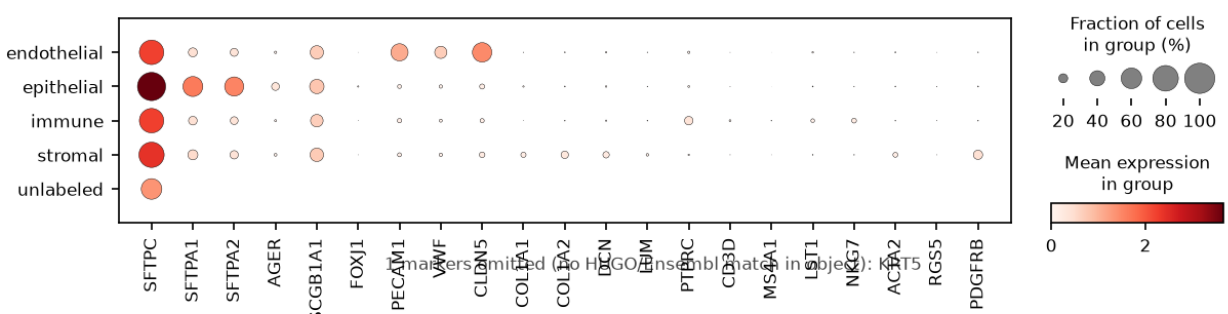


Side-by-side UMAP comparison of label views after Harmony

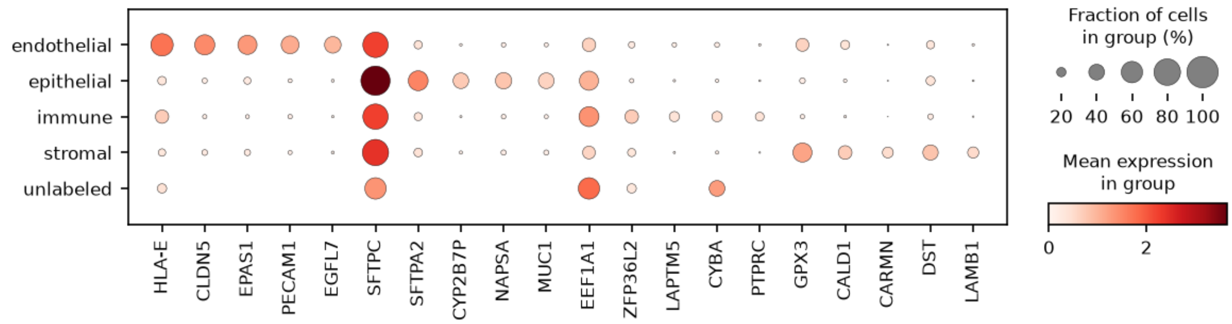
Composition versus markers



Mean expression heatmap of known lung markers across coarse cell classes



Dotplot of known lung marker genes by cell_class

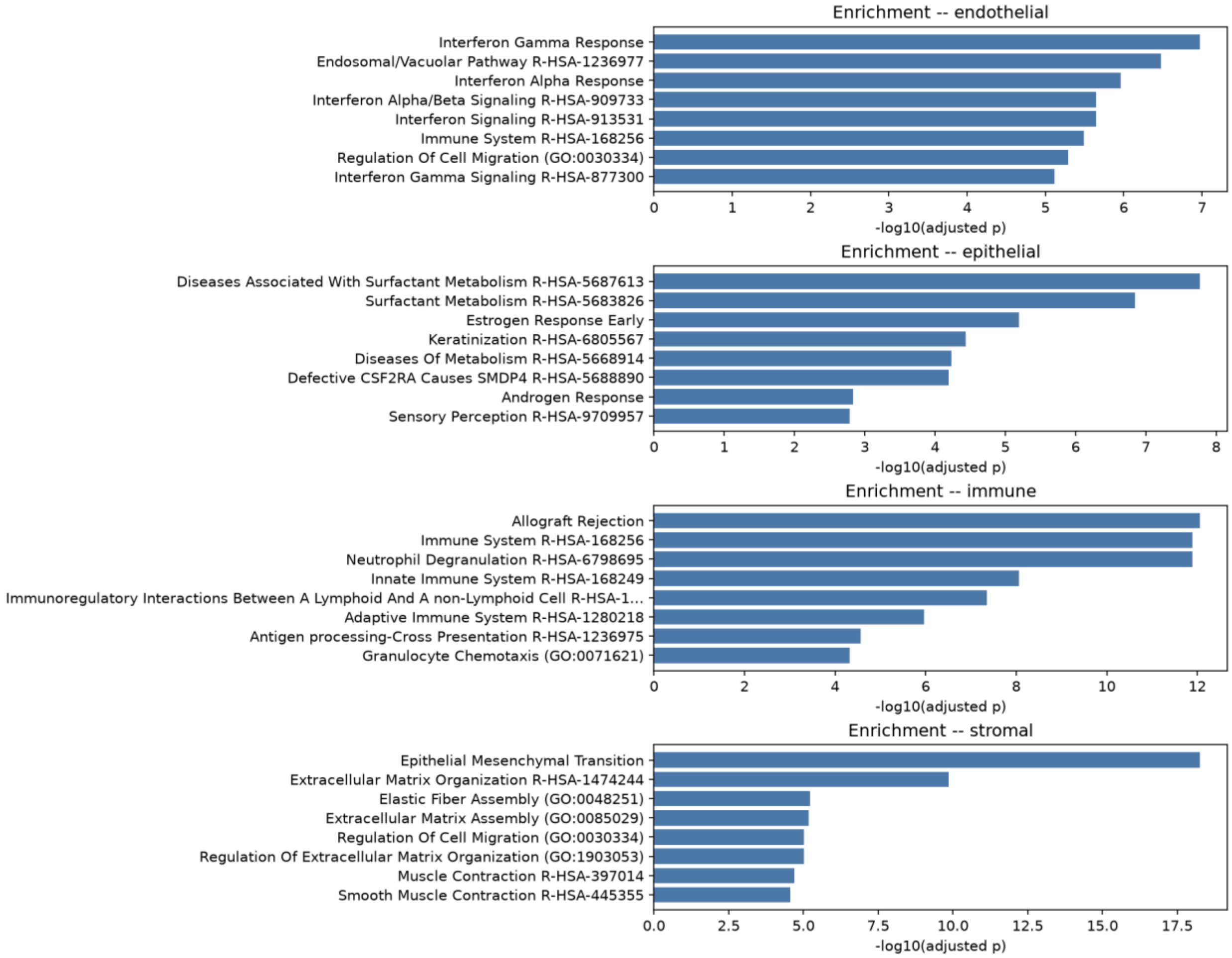


Dotplot of top Wilcoxon markers by cell_class

Wilcoxon ranking returns 9,921 rows. Under the module gates, adjusted p below 0.05 and log fold change above 0.25, 907 epithelial, 456 endothelial, 185 stromal, and 139 immune markers survive. Top ranks are lineage-appropriate but uneven: CLDN5 leads endothelial at log fold change 4.06, detected in 55.4 percent of endothelial nuclei versus 9.1 percent elsewhere, while SFTPC leads epithelial at 2.10 yet appears in 76.2 percent of the non-epithelial reference, which is ambient surfactant. Of 22 panel genes, 13 resolve by symbol and 21 by Ensembl ID; only KRT5 is missing from both.

Local ORA on 100 input genes per class returns 724 terms at adjusted p below 0.05 (endothelial 259, stromal 238, immune 190, epithelial 37); prerank GSEA returns 225 at FDR below 0.05. Some read as biology: Epithelial Mesenchymal Transition tops stromal at 19 of 143 genes. Two do not. Hallmark Estrogen Response Early is a top epithelial term at 9 of 143, adjusted p 6.5e-06, and Hallmark Allograft Rejection tops immune at 14 of 127, adjusted p 8.7e-13, in lung with no transplant anywhere. Both are real output of this run and both are gene-overlap artifacts.

Full tables: outputs/tables/module2_marker_genes.tsv, module2_pathway_enrichment.tsv, module2_prerank_gsea.tsv, module2_celltype_fractions.tsv, module2_donor_by_label_crosstab.tsv.



Summary of local GMT enrichment results by cell_class

5. Interpretation

Module 2 shows that the four QC'd HuBMAP donors can be embedded jointly, that Harmony increases donor mixing by a measurable neighborhood metric, and that coarse lineages yield markers and enrichment tables suitable for classroom interpretation.

What the results support.

- Integration is usable for teaching composition and marker analyses within one DCC.
- Composition tables and marker tables answer different questions; both are required to avoid attributing abundance shifts to cell-state programs.
- Foreign-donor kNN fraction is a concrete mixing statistic learners can recompute after changing Harmony or PCA settings.

How to read markers and enrichment together. A large epithelial compartment will dominate many global summaries. Class-stratified Wilcoxon markers ask what distinguishes each coarse class inside the joint object. Enrichment then summarizes those ranked lists against gene sets. When n_input_genes is small, ORA terms become brittle: the same course packages Hallmark (broad) and Reactome/GO BP (finer). Prefer terms that recur across libraries and that match known lung marker biology in the dotplots before elevating a pathway to a claim.

6. Extending this analysis

Each item below states what a stronger version of this analysis would require. The requirement is the boundary of what the Module 2 tables currently license.

- **Cross-ecosystem agreement needs a second ecosystem.** Establishing that these lineage programs recur in GTEx, GEO, or spaceflight tissue requires a second source measured independently and a bridge between the two feature spaces. Module 4 builds exactly that from the composition-weighted pseudobulk exported here; until then every number in this module comes from four HuBMAP donors and describes only them.
- **Confirming an enriched pathway requires evidence outside the ranked list.** The standard is term-by-term inspection of gene membership, recurrence across Hallmark and Reactome and GO Biological Process, and an orthogonal assay of the implicated program. Estrogen Response Early and Allograft Rejection both clear the adjusted p gate in this run and neither would survive that standard, which is why the check is run rather than assumed.
- **A donor-age contrast that carries weight requires donors, not a split.** The n=2 vs n=2 comparison demonstrates the mechanics of setting up a covariate contrast. Powering it would require a cohort sized against between-donor variance, which is a cohort design problem rather than a parameter change.
- **Showing that Harmony removed batch, rather than rearranged the embedding, would require a matched re-prep.** The same donor processed across protocols is what separates donor identity from technical batch. Foreign-donor kNN fraction and within-class donor silhouette are the two metrics this module chose to monitor; they are diagnostic of the embedding, not causal about its source.
- **Trusting a mixing verdict per compartment requires per-compartment reporting.** The immune silhouette moving toward zero while three compartments move negative is visible only because the silhouette table is stratified by cell_class. A single global mixing number would require a stratified companion metric before anyone reads it as uniform improvement.

Forward link. Module 3 interprets paired RNA+ATAC for Donor_2 (HBM828.GPVG.252), matched to the Donor_2 10x snRNA teaching block. Module 4 reuses composition-weighted pseudobulk in the HuBMAP-GTEx bridge and compares cross-ecosystem contrasts.

7. Open questions

1. Which cell population or cluster was analyzed?
2. What are the top marker genes, and do they match known lung identities?
3. What pathways are enriched, and are they biologically coherent for that population?
4. What biological interpretation is supported by markers + pathways together?
5. What limitations remain (label transfer uncertainty, enrichment bias, small input lists)?
6. Did Harmony change donor mixing in a way that matches the silhouette story, or do the two metrics disagree?
7. Where does composition change stop and cell-state change begin in these tables?

Appendix A: Analysis parameters

Live values under module2 in config/paths.yaml. Section numbers match the Methods subsections.

1. Load and concatenate Module 1 QC objects

Config key	Default	Definition	If modified	Impact on results
module2.inputs	four donor blocks	Cohort of Module 1 h5ads to concat	Drop/add donor	Every composition fraction and DE n changes
module2.input_h5ad	data/processed/integrated/module2_hubmap_lung_integrated.h5ad	Written integrated path (DE input)	Relocate	Downstream DE must point at new path
module1.max_nuclei	5000	Upstream teaching cap (per donor)	Raise/lower	Nucleus counts entering concat
module2.random_seed	0	Seed for later embedding / diagnostics	Change	Stochastic neighbors/UMAP/Harmony draws

2. Joint embedding (recomputed HVG / PCA)

Config key	Default	Definition	If modified	Impact on results
module2.embedding.n_top_genes	2000	Joint HVG count (Seurat flavor)	Raise/lower	Changes PCA basis and all neighbors
module2.embedding.n_pcs	40	PCA components recomputed jointly	Raise	Can reintroduce donor-driven variance into neighbors
module2.embedding.n_pcs_neighbors	40	PCs used for kNN	Lower	Smoother neighborhoods; may lose fine structure
module2.embedding.n_neighbors	20	kNN size	Raise	More global mixing in UMAP
module2.embedding.umap_min_dist	0.25	UMAP min_dist	Lower	Tighter visual clusters
module2.embedding.umap_spread	1.2	UMAP spread	Raise	More dispersed layout
module2.embedding.scale_max_value	10	Scale clip before PCA	Lower	Compresses outlier gene leverage
module2.embedding.leiden_resolution	0.8	Default Leiden for teaching	Change	Cluster grain; default label still Azimuth-driven
module2.embedding.leiden_resolutions	[0.4, 0.8, 1.2]	Exploratory sweep stored as leiden_r*	Edit list	Extra cluster columns only

3. Harmony batch correction

Config key	Default	Definition	If modified	Impact on results
module2.harmony.enabled	true	Whether Harmony runs	Set false	No X_pca_harmony; mixing comparison collapses
module2.harmony.key	donor_id	Obs column for batch	Use donor_label	Same biological batches if IDs align
module2.harmony.max_iter_harmony	20	Harmony iterations	Raise	Longer runtime; usually small geometry change
module2.harmony.theta	null	Diversity penalty (null = package default)	Raise	Stronger correction; risk of overmixing rare states
module2.random_seed	0	Passed to Harmony / neighbors	Change	Stochastic differences in corrected space

4. Mixing and silhouette diagnostics

Config key	Default	Definition	If modified	Impact on results
module2.diagnostics.enabled	true	Run mixing / silhouette	Set false	No mixing tables; RAM floor may omit diagnostic peak
module2.diagnostics.log_matrix_memory	true	Log matrix RSS notes	Set false	Reporting only
module2.diagnostics.knn_k	30	Neighbors for foreign-donor fraction	Change k	Numeric mixing score scale changes
module2.diagnostics.max_cells	8000	Subsample cap for scoring	Raise	Higher RAM; stabler estimate
module2.diagnostics.random_seed	null	Null falls back to module2.random_seed	Set explicit	Controls diagnostic subsample only

5. Coarse cell_class mapping, composition, and pseudobulk

Config key	Default	Definition	If modified	Impact on results
module2.label_column	azimuth_label	Fine label source for mapping	Switch column	Remaps every cell_class
module2.labels.fetch_remote	false	Remote range-read for labels	Set true	Join coverage may change
module2.labels.cache	outputs/tables/module2_azimuth_labels.tsv	Local label cache	Refresh	Same as join change
module2.groupby	cell_class	Default DE grouping	Use fine labels	Marker tables explode in groups
module2.export_pseudobulk	true	Write bridge vectors for Module 4	Set false	Module 4 level bridge lacks HuBMAP side
module2.markers	lung panel list	Genes for teaching heatmaps/dotplots	Edit list	Figure content only

module2.known_markers	lineage panels	Named marker sets for dotplots	Edit	Figure content only

6. Intron exclusion on the DE path, Wilcoxon markers

Config key	Default	Definition	If modified	Impact on results
module2.exclude_intron_features	true	Drop *-I features at DE load	Set false	Intron tags compete in ranks; markers change
module2.de.method	wilcoxon	rank_genes_groups method	Change	Rank statistics differ
module2.de.n_genes	3000	Genes reported per group (prerank-friendly)	Lower	Shorter ranked lists for GSEA
module2.de.min_cells_per_group	50	Minimum nuclei to keep a group	Raise	Rare classes may drop from DE
module2.de.min_logfoldchange	0.25	Filter for ORA input markers	Raise	Fewer ORA genes; brittle enrichment
module2.de.max_padj	0.05	Adjusted p filter for markers	Stricter	Same direction as LFC filter
module2.de.exclude_ribosomal_genes	true	Drop ribosomal symbols from marker tables	Set false	Ribosomal genes dominate ranks
module2.azimuth_focus	AT2/AT1/...	Fine labels highlighted in plots	Edit	Figure focus only

7. Local GMT enrichment (ORA and prerank)

Config key	Default	Definition	If modified	Impact on results
module2.enrichment.enabled	true	Run enrichment half	Set false	No pathway tables
module2.enrichment.mode	local_gmt	Offline GMT vs Enrichr HTTP	enrichr_online	Needs network; different background
module2.enrichment.gene_sets	Hallmark, Reactome, GO BP	GMT basenames under data/genesets	Swap release	Term names/sizes change
module2.enrichment.geneset_dir	data/genesets	GMT directory	Relocate	Path resolution
module2.enrichment.top_genes_per_group	100	ORA input list size	Lower	Unstable ORA
module2.enrichment.cutoff_padj	0.05	ORA marker padj gate	Stricter	Fewer input genes
module2.enrichment.prerank.enabled	true	Run prerank GSEA	Set false	No NES table for Module 2
module2.enrichment.prerank.min_size	15	Min overlapping genes in set	Raise	Drops small sets
module2.enrichment.prerank.max_size	500	Max set size	Lower	Drops large sets
module2.enrichment.prerank.permutation_num	100	Prerank permutations	Raise	Stabler FDR; slower

module2.enrichment.exclude_ribosomal_genes	null	Null inherits DE ribosomal policy	Override	Changes ranked inputs

References

8. Korsunsky I, et al. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat Methods*. 2019;16(12):1289-1296. PMID: 31740819.
9. Subramanian A, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A*. 2005;102(43):15545-15550. PMID: 16199517.
10. Wolf FA, Angerer P, Theis FJ. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol*. 2018;19(1):15. PMID: 29409532.
11. Liberzon A, et al. The Molecular Signatures Database (MSigDB) hallmark gene set collection. *Cell Syst*. 2015;1(6):417-425. PMID: 26771021.